

Bayesian Model Averaging

454 Capstone Project

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01

ABSTRACT

ABSTRACT

The Scenario : Can We Predict Final Exam Scores?

- The Goal (y): Predict a student's final exam grade.
- The predictors: Study Hours; Library Check-Ins; Assignment Completion; Extracurriculars; Having Breakfast...



02

INTRODUCTION

What is Bayesian Model Averaging?

- **Definition:** Instead of picking one model, BMA takes a weighted average of **all** possible models
- The Math:

$$P(\Delta \mid D) = \sum_{k=1}^K P(\Delta \mid M_k, D) P(M_k \mid D)$$


Prediction
from a single
model

Model Weights

The Scenario

- The Goal (y): Predict a student's final exam grade.
- The predictors:
 - Study Hours (x_1)
 - Library Check-Ins (x_2)
 - Assignment Completion (x_3)
 - Lucky Socks (x_4)

$$\text{Final Prediction} = w_1(x_1) + w_2(x_1 + x_2) + w_3(x_1 + x_3) + \dots$$

Shifting the Research Mindset

	Traditional Approach	BMA Approach
Question	Does Study Time have a statistically significant relationship with Grade?	What factors and combination best helps us understand and predict the Grade?
Assumption	Predefined the 'correct' model	Uncertain which model is true
Standard	P-value	Posterior Inclusion Probability (PIP)
Risk	Ignoring model uncertainty	Accounts for model uncertainty

Why BMA?

- Model Uncertainty
- Robust
- Neutral filter

However...

The Problem...

# predictors (p)	# models ($K = 2^p$)
4	16
5	32
6	64
7	128
8	256
10	1,024
15	32,768
20	1,048,576
50	1.1259×10^{15}

BMA:

$$P(\Delta \mid D) = \sum_{k=1}^K P(\Delta \mid M_k, D) P(M_k \mid D)$$

- BMA requires averaging over *all* possible models, K
- Most models have very low posterior probability
 - $P(M_k \mid D) \approx 0$

← one quadrillion!

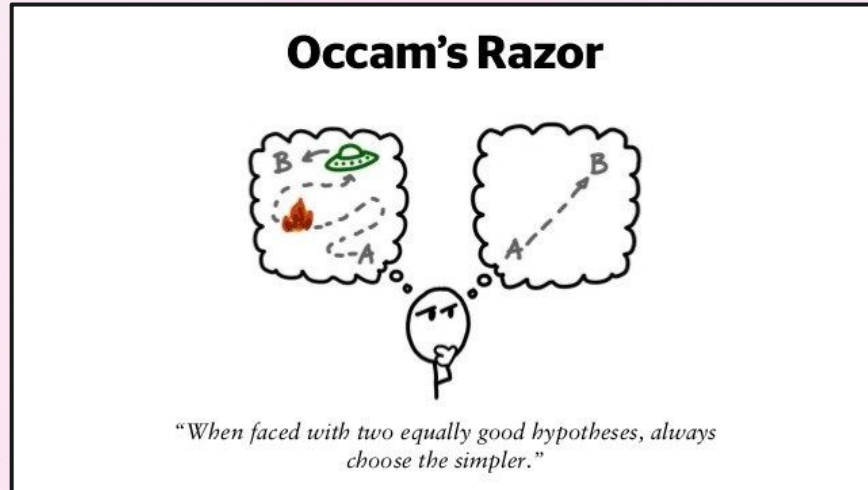
03

OCCAM'S WINDOW

Occam's Window

Filter down candidate models to keep “plausible” models

- Compare each model to the “best” model (highest posterior probability)
- Remove models that are too unlikely
- Also remove overly complex models dominated by simpler ones



An (Occam's) Window into the Math

Let M^* be the model with highest posterior probability.

Then only keep models that satisfy

$$\frac{P(M_k \mid D)}{P(M^* \mid D)} > \frac{1}{C}$$



- C is threshold (commonly 20-100 (Madigan & Raftery 1994))
- # of models considered significantly reduced
- All other models are “ignored”

OPTIONAL:

Let M_{complex} be nested model containing M_{simple} , $M_{\text{simple}} \subset M_{\text{complex}}$

Remove M_{complex} if $P(M_{\text{simple}}) > P(M_{\text{complex}})$

Algorithm

Candidate Models

Fit all possible models

p predictors, 2^p models

Posterior Probabilities

Probability that each model is correct given the data

Better fit = higher value

Best Model

Model with highest posterior probability

Used as reference for comparison

Occam's Window

Remove models that are **much** less likely than the best model

$$\frac{P(M_k | D)}{P(M_* | D)} > \frac{1}{C}$$

Optionally,

$$P(M_{\text{simple}}) > P(M_{\text{complex}})$$

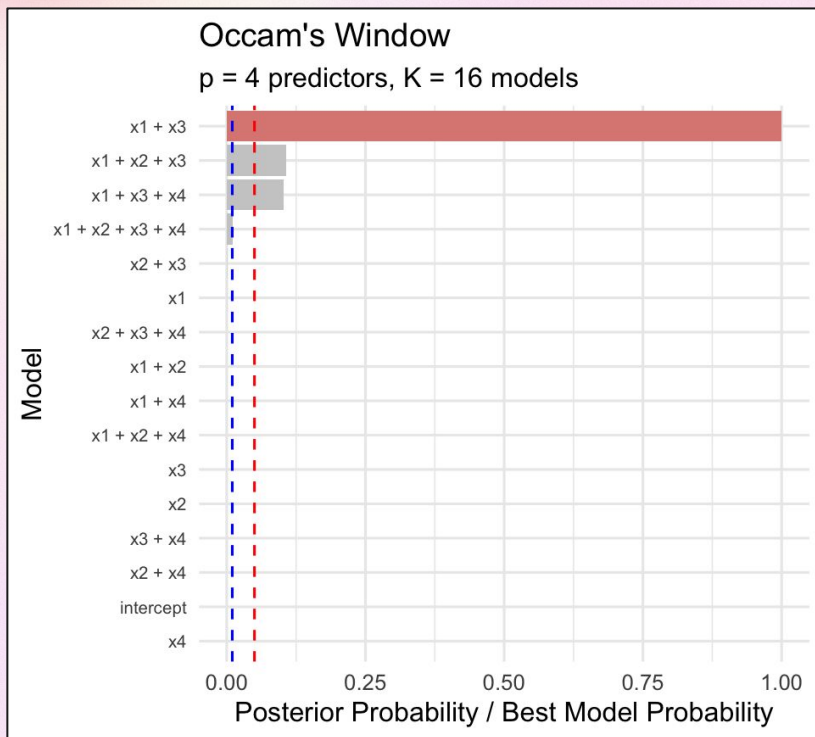
$$M_{\text{simple}} \subset M_{\text{complex}}$$

BMA

Average predictions across the remaining models using their posterior weights

Results account for model uncertainty

Simple Example



$$\frac{P(M_k | D)}{P(M^* | D)} > \frac{1}{C}$$

$$M^* = x_1 + x_3$$

- K = 16 reduced to
 - 3 models
 - 4 models

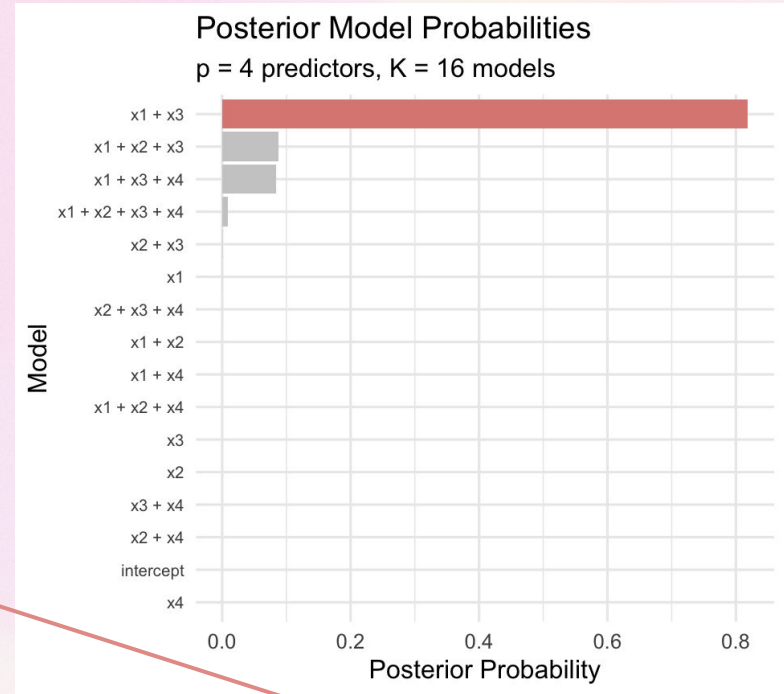
OW Cutoffs: $\frac{1}{20}$
 $\frac{1}{100}$

Why Occam's Window Works

- Most posterior mass is concentrated in a small subset of models
- Weak models contribute negligibly to predictions
- Complex models are only kept if they provide substantially better fit than simpler ones

BMA:

$$P(\Delta \mid D) = \sum_{k=1}^K P(\Delta \mid M_k, D) P(M_k \mid D)$$



$P(M_k \mid D)$

04

SIMULATED DATA

Data Generation

$$y_i = 2 + 2x_{1i} + 0.8x_{3i} + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, 2.5^2)$$

$$x_{1i}, x_{3i} \stackrel{\text{iid}}{\sim} \text{Normal}(0, 1)$$

$$x_{2i} = 0.85x_{1i} + \alpha_i, \quad \alpha_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, 0.5^2)$$

$$x_{4i}, \dots, x_{15i} \stackrel{\text{iid}}{\sim} \text{Normal}(0, 1)$$

- $i = 1, \dots, 100$ observations
- $x_1, x_3 \rightarrow$ “true” predictors of y
- $x_2 \rightarrow$ correlated with x_1 plus noise
- $x_4, \dots, x_{15} \rightarrow$ random noise

05

APPROXIMATING BMA

Posterior Inclusion Probability (PIP)

The model-averaged probability of including a certain predictor (x_j) in the model, given the observations

$$\text{PIP}(x_j) = P(x_j \text{ is in the model} \mid D)$$

How important is each variable across *all* plausible models?

Simple MH MCMC for Model Selection

- Consider $x_1, \dots, x_4$, y is generated from x_1 and x_3 + random noise (“truth”)
- Goal: recover important variables without knowing the true model

Each iteration of Markov chain is a model:

ex. $(1, 0, 1, 0) \rightarrow y \sim x_1 + x_3$

$(1, 1, 1, 1) \rightarrow y \sim x_1 + x_2 + x_3 + x_4$

$(0, 0, 0, 0) \rightarrow y \sim 1$

chain				
	[,1]	[,2]	[,3]	[,4]
[1,]	0	0	0	0
[2,]	1	0	0	0
[3,]	1	0	1	0
[4,]	1	0	1	0
[5,]	1	0	1	0
[6,]	1	0	1	0
[7,]	1	0	1	0
[8,]	1	0	1	0
[9,]	1	0	1	0
[10,]	1	0	1	0
[11,]	1	0	1	0
[12,]	1	0	1	0
[13,]	1	0	1	1

Simple MH MCMC - Algorithm Outline

Each step:

- Proposes new model by flipping one value, ex. $(1, 1, 1, 1) \rightarrow (0, 1, 1, 1)$
- Approximate log posterior model probability:
 - $-0.5 * \text{BIC}(\text{model})$ (Hoeting et al. 1999)
- Always accept better models; accept worse proposed models with some probability to ensure exploration

$$\alpha = \min \left(1, \frac{P(M_{\text{proposed}} \mid D)}{P(M_{\text{current}} \mid D)} \right)$$
$$= \min(1, \exp(-\frac{1}{2}[BIC_{\text{proposed}} - BIC_{\text{current}}]))$$

PIP:

- How frequently does the algorithm include models with that predictor?


```

set.seed(123)
p <- 4 # number of predictors
# start at a model (x1 and x3 included)
gamma_current <- c(1, 0, 1, 0)
n_iter <- 5000
chain <- matrix(NA, n_iter, p) # 5000 row, 4 column matrix
chain[1, ] <- gamma_current # set first value
# posterior score
log_post <- function(gamma){ # gamma = binary vector
  vars <- c("x1","x2","x3","x4")[gamma == 1] # translate to variable names for lm formula
  if (length(vars) == 0) {
    fit <- lm(y ~ 1, data = sim_data) # intercept model
  } else {
    form <- as.formula(paste("y ~", paste(vars, collapse = " + ")))
    fit <- lm(form, data = sim_data)
  }
  -0.5 * BIC(fit) # logP(Mk | D) is approx -.5*BIC, Bayesian approximation
}
# MARKOV CHAIN!
for (i in 2:n_iter) {
  # propose: flip one variable
  gamma_prop <- gamma_current
  j <- sample(1:p, 1) # random number 1-p
  gamma_prop[j] <- 1 - gamma_prop[j] # 1 - 1(currently included) = 0 (excluded), 1 - 0 (currently excluded) = 1 (included)
  # acceptance probability, Is this new model more likely (posterior probability) than the current model?
  alpha <- min(1, exp(log_post(gamma_prop) - log_post(gamma_current)))
  # accept/reject
  if (runif(1) < alpha) {
    gamma_current <- gamma_prop # accept proposal
  }
  chain[i, ] <- gamma_current # update chain
}

```

```

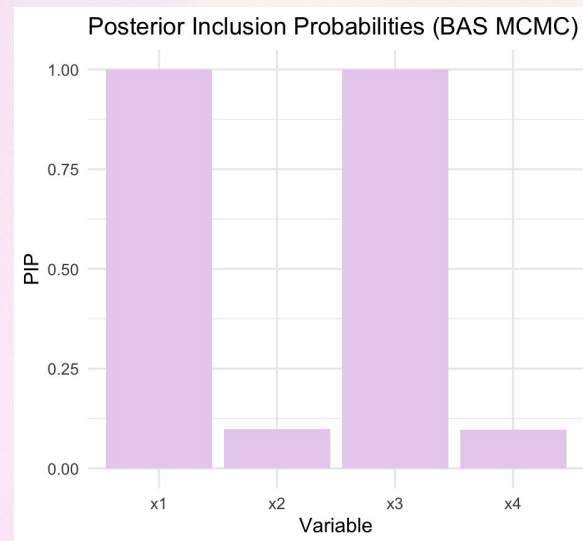
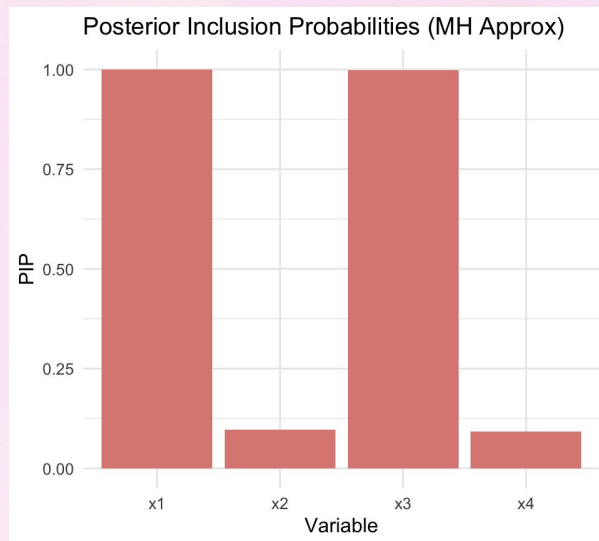
library(BAS)
set.seed(123)
bas_fit <- bas.lm(y ~ x1 + x2 + x3 + x4,
  data = sim_data,
  method = "MCMC",
  MCMC.iterations = 5000,
  prior = "BIC",
  modelprior = uniform())

```

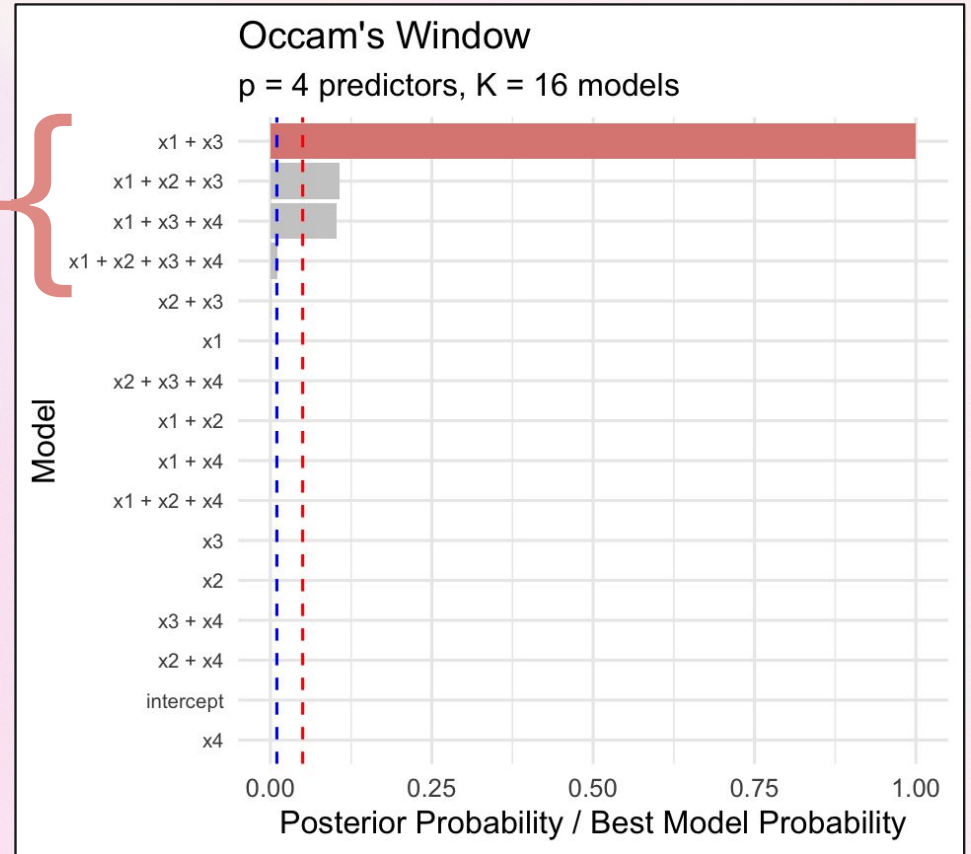
models not weighed
differently in MH
MCMC

MCMC vs BAS Package (simple)

Variable	PIP (MH)	PIP (BAS)
x1	0.9996	0.9992
x2	0.0966	0.0983
x3	0.9986	0.9994
x4	0.0920	0.0969



Keeps only models with
x1 and **x3** together



BAS R Package

- If # of predictors < 25, BAS uses all models ($2^{25} = 33,554,432$ models)
- When > 25 predictors...
 - Bayesian Adaptive Sampling (BAS)
 - Adaptive MCMC
 - Combination of the two algorithms
- Maintained by Merlise Clyde (Duke)



06

EXPANSION

Expanding the Model

Baseline: 4 variables $\rightarrow 2^4 = 16$ models



Expansion: Adding 11 noise variables (noise₁ to noise₁₁)



After Expansion: 15 predictors $\rightarrow 2^{15} = 32,768$ models

* The Limit: computationally expensive and slow

The Solution: M-H MCMC

```
```{r}
set.seed(454)
n <- 100 #sample size

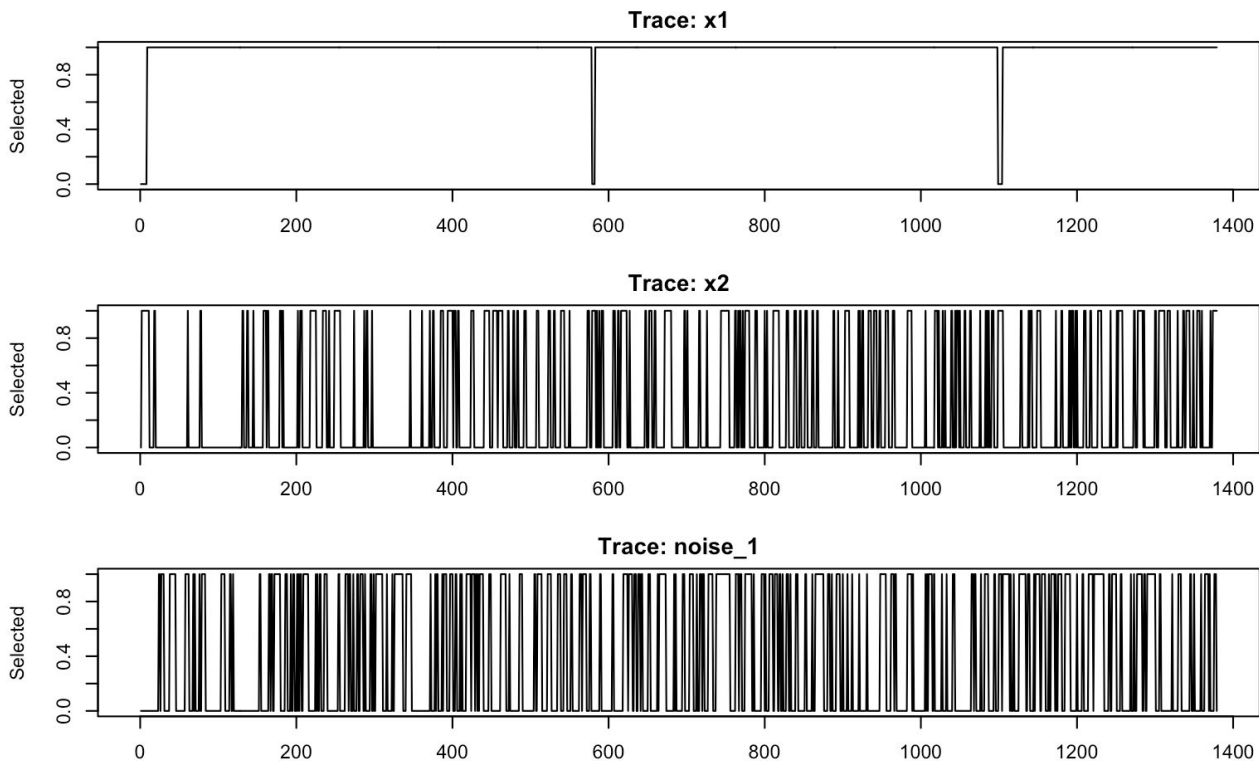
x1 <- rnorm(n); x2 <- 0.85*x1 + rnorm(n,0,0.5); x3 <- rnorm(n); x4 <- rnorm(n)

creating 11 noise predictors
noise_vars <- matrix(rnorm(n * 11), ncol = 11)
colnames(noise_vars) <- paste0("noise_", 1:11)
y = 2 + 2*x1 + 0.8*x3 + rnorm(n, 0, 2.5)

#combine all 15 predictors
sim_data_15 <- data.frame(y, x1, x2, x3, x4, noise_vars)

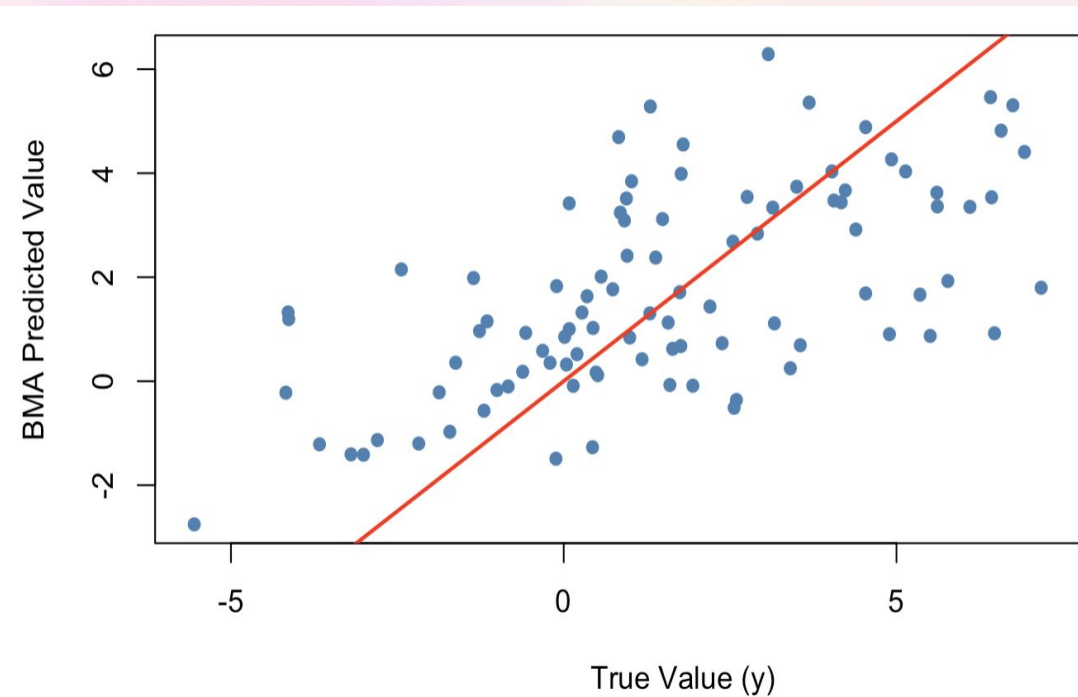
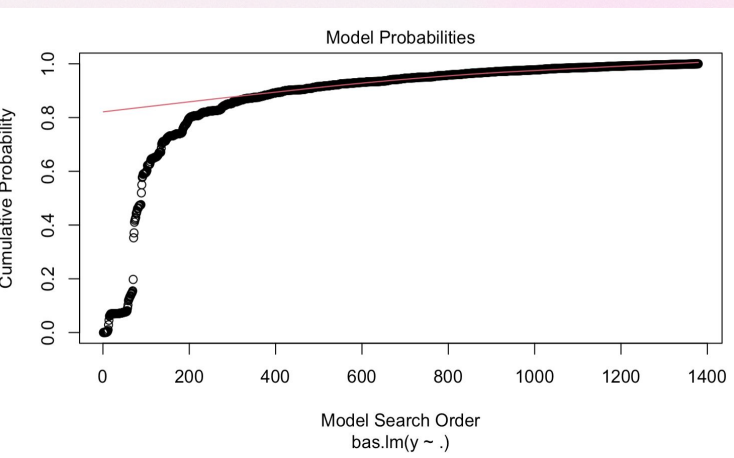
Use BAS package for MCMC
library(BAS)
bas_15 <- bas.lm(y ~ ., # "."
 data = sim_data_15,
 prior = "BIC",
 modelprior = uniform(),
 method = "MCMC",
 MCMC.iterations = 50000) # more iterations because more predictors
```

# MCMC Convergence





# Diagnostics







# References

Amini, S. M., & Parmeter, C. F. (2011). Bayesian model averaging in R. *Journal of Economic and Social Measurement*, 36(4), 253-287.

Hoeting, J. A., Madigan, D., Raftery, A. E., & Volinsky, C. T. (1999). Bayesian model averaging: a tutorial (with comments by M. Clyde, David Draper and El George, and a rejoinder by the authors). *Statistical science*, 14(4), 382-417.

Madigan, D., & Raftery, A. E. (1994). Model Selection and Accounting for Model Uncertainty in Graphical Models Using Occam's Window. *Journal of the American Statistical Association*, 89(428), 1535-1546.  
<https://doi.org/10.2307/2291017>

**THANKS!**